Sahaj Bhadja

(213) 284-4651 bhadja@usc.edu sahaj-bhadja sahajbhadja sahaj-bhadja

Summary

Software Engineer with 2+ years building scalable web applications and backend services using **C#, .NET, Python, SQL Server, REST APIs**, asynchronous processing, React, cloud infrastructure, CI/CD, and automated testing.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4*
Coursework: Database Systems, Web Technologies, Software Engineering, Analysis of Algorithms

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4*
Coursework: Object-Oriented Programming, Software Engineering, Data Retrieval, Operating Systems

Experience

USC Dornsife **September 2025 – Present**
Software Engineer *LA, California*

- Designed and optimized a multi-tenant PostgreSQL and SQL Server data model supporting 200+ users and 10K+ records, reducing application latency 35% through indexing, validation, and query tuning.
- Restructured fault-tolerant asynchronous orchestration with explicit state transitions and failure isolation, processing 1K+ jobs while tripling throughput under concurrent workloads and preventing cascading failures across dependent services.

Grad8 **June 2025 – August 2025**
Software Engineer Intern *LA, California*

- Designed concurrent backend workflows using multithreading and secure file validation, processing 1K+ files per batch through AWS RDS and REST APIs with isolated, observable, recoverable, fault-tolerant execution.
- Implemented production services handling 10K+ daily requests for 150+ users, applying CI/CD-backed testing and deployment workflows to reduce turnaround 60% while maintaining consistently stable application peak-load performance.

USC AD Labs **January 2025 – May 2025**
ML Research Assistant *Los Angeles*

- Built a Python simulation pipeline processing 100K+ trajectory logs, improving long-horizon planning 18% while rigorously testing data transformations and system behavior across repeatable, controlled benchmark evaluation scenarios.
- Developed a spatial-temporal encoder combining map geometry and trajectory context, improving closed-loop stability 15% through systematic regression testing, detailed failure analysis, structured root-cause debugging, and documented evaluation.

Aasma Technology Solutions **August 2023 – July 2024**
Software Developer Engineer - (SecureSync) *India*

- Engineered cross-platform microservices with automated VM builds and 20+ robust unit and integration suites using Jenkins, PyTest, and Mockito across Windows, Linux, and macOS production release environments.
- Optimized object-oriented API integrations and AWS data flows, improving response times 30%, pipeline efficiency 68%, and maintainability through reusable Factory and Singleton patterns and structured code reviews.

Aasma Technology Solutions **May 2023 – August 2023**
Full Stack Developer Intern - (VibrantChem) *India*

- Developed a MERN business platform using React, Node.js, REST APIs, WebSockets, and eight modular services, synchronizing 2M+ records across distributed locations while reducing code-review turnaround by 25%.
- Automated Docker-based application deployments through Jenkins CI/CD and integrated test validation, cutting deployment time 50% while achieving 95%+ successful build validation across repeated production software release cycles.

Skills

Languages: C#, Python, Java, JavaScript, TypeScript, C++, SQL, HTML, CSS, Bash
Backend & Security: .NET, Async/Await, OOP, REST APIs, Web APIs, Authentication, Authorization, Session Management, Microservices
Frontend & Data: React, Node.js, SQL Server, PostgreSQL, MySQL, MongoDB, Database Design, Data Modeling
Cloud & Quality: Azure, AWS, Docker, Kubernetes, Git, CI/CD, Jenkins, Unit Testing, Integration Testing, Debugging, Code Reviews, Agile

Projects

BookingCore | C#, .NET, Async/Await, Authentication, Unit Testing

- Built and compiled an asynchronous C#/.NET booking component with role-based sessions, idempotent requests, provider collision prevention, tenant-isolated queries, input validation, and five executable automated behavioral reliability tests.

HMIS - Hospital Management Information System Live | TypeScript, REST APIs, Monitoring

- Architected a service-based application integrating ABDM APIs, supporting 100+ daily appointment workflows and reducing record retrieval time by 2x through structured schemas, consent controls, and production monitoring.

FinNexus Live | Next.js, TypeScript, MongoDB, Kubernetes

- Built a secure fintech platform supporting 50K+ users and 1M+ transactions, integrating Plaid bank linking, token-based request controls, real-time synchronization, advanced filtering, and reliable containerized deployment workflows.